

Data Polishing — Clean Messy CSV

Day 08

Lesson 08

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Course 02

Phase 2 — Data Lab

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Spot bad rows: empty cells, wrong types, duplicates
- ★ Write `clean_fights.csv` with fixed rows
- ★ Document what you fixed



Course 02 quest map: Day 08 of 18

Hook

Real data is messy. Analysts spend time cleaning before charting.

Learn

Learn

Common problems:

- empty `command`
- `player_hp` text instead of number
- duplicate round numbers
- extra spaces: " fight "

Fix with `strip()`, `int()`, skip bad rows, or merge duplicates.

AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — **Run** — **Investigate** — **Modify** → **Make**

Demo code

Folder: `day08/`

Open: `day08/demos/messy_fights.csv`

 Try it

Open `demos/messy_fights.csv`. List 3 problems you see **before** writing code.

 Assignment

Work in: `day08/assignment/`

A — Written List 3 data problems in the messy file (no code).

B — Practice Run `demos/clean_demo.py` and read how it fixes rows.

C — Level up `assignment/clean_fights.py` reads `assignment/messy_source.csv`, writes `clean_fights.csv`.

D — Explain `assignment/CLEANING_LOG.md`: what you fixed and how many rows removed.

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Compute win rate and average damage.